

CPRE 4880 - Section A1

MP - 2

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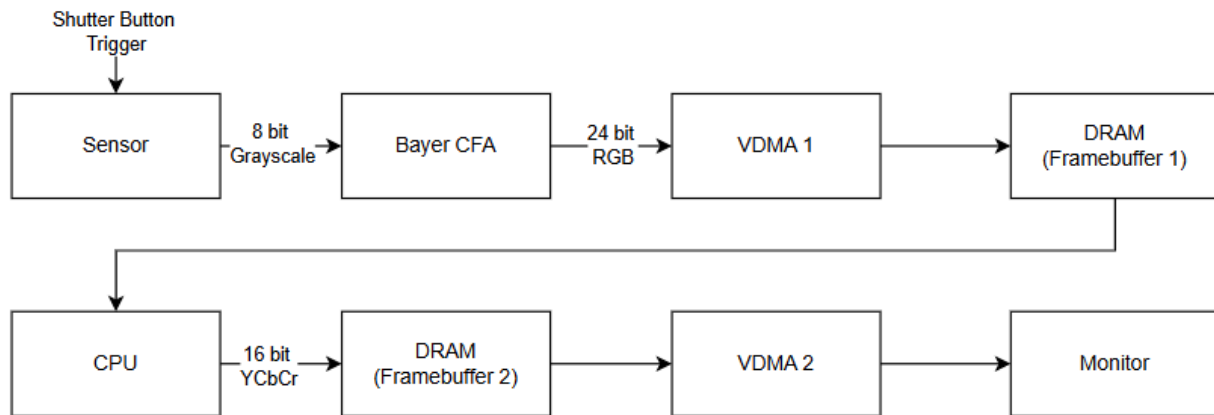
Dr. Phillip Jones

Table of Contents

Design Test and Analysis	3
A detailed system diagram that illustrates the interconnection between the various modules in the system	3
A description of how the hardware in the starter MP-2 design is intended to operate	3
Describe in your write-up what changes you made, and save a copy of any files modified (presumably only camera_app.cc) during this process into a folder named part3	3
Grayscale Camera	4
Describe what this pairing of signals signifies, and what this configuration is typically used for	4
Explain why this is an appropriate value to append, and why appending "00000000" would not make sense	4
Briefly explain why the camera at this stage is not outputting any color	4
Color Conversion Software	5
Use this format as the output of your camera_loop() conversion pass	5
Describe the performance of your software-based color conversion (in terms of frames per second)	6
Describe your testing methodology	6
Image Processing Pipeline	7
Include a diagram of this pipeline in your write-up, labeling the bit widths of the relevant signals	7
Describe the performance of your image processing pipeline (in terms of frames per second), and how you measured it	7

Design Test and Analysis.

A detailed system diagram that illustrates the interconnection between the various modules in the system

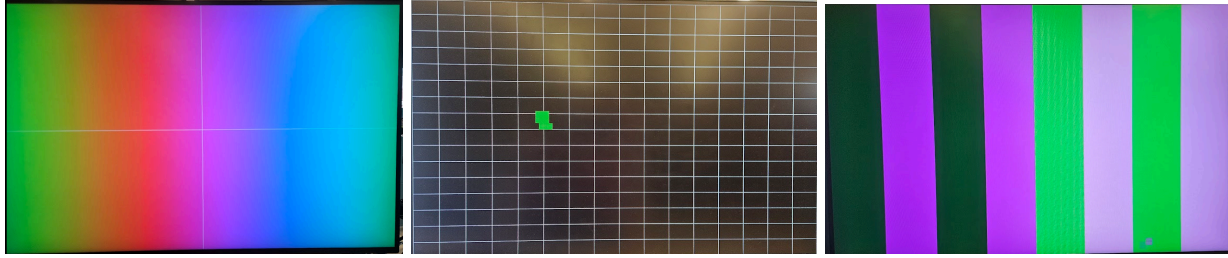


A description of how the hardware in the starter MP-2 design is intended to operate.

1. Uses the Processing System (PS) to configure everything.
2. Generates video data in the FPGA.
3. Moves video frames to/from DDR memory using AXI VDMA.
4. Converts AXI-Stream video into HDMI signals.
5. Outputs HDMI video to a monitor.

Describe in your write-up what changes you made, and save a copy of any files modified (presumably only camera_app.cc) during this process into a folder named part3.

Changing the registers, we got the following images:



By changing the software loop, we made the image appear upright instead of upside down.
`pMM2S_Mem[i] = pS2MM_Mem[i]; //1920*1080-i-1`

Grayscale Camera

Describe what this pairing of signals signifies, and what this configuration is typically used for

The different suffixes on the signals are the differential pairings. The positive (`_p`) carries the true signal, and the negative (`_n`) carries the logical inverse. Differential pairs are used in high-speed digital transmission because the voltage swing is small, allowing them to switch quickly without radiating much EMI.

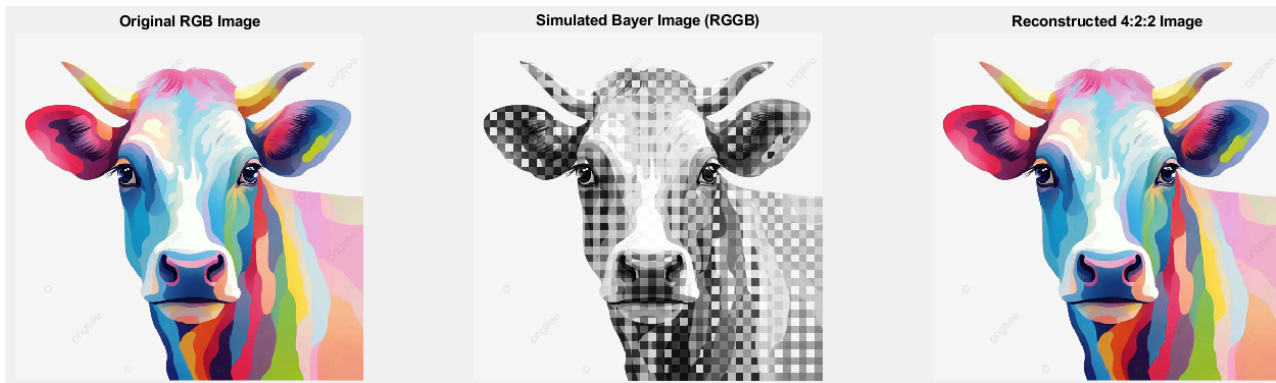
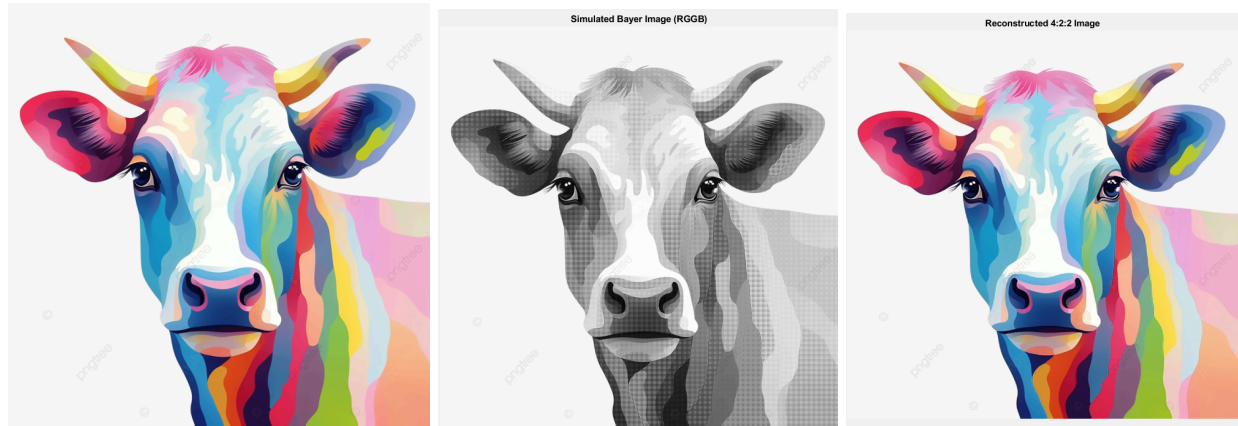
Explain why this is an appropriate value to append, and why appending “00000000” would not make sense.

We append “10000000” because we want a data width of 128; if we append “00000000”, we might get a data width of 0, meaning no data is being transmitted.

Briefly explain why the camera at this stage is not outputting any color

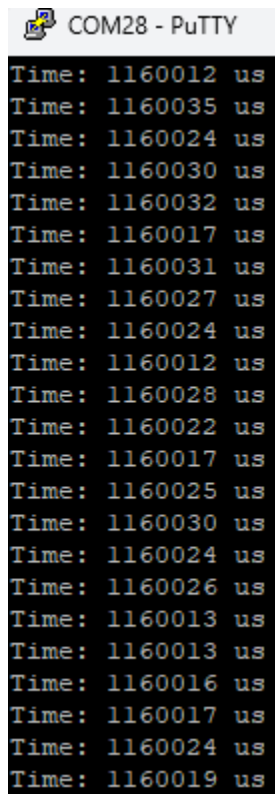
The camera sensor outputs Bayer pattern data, and each pixel has only one color channel. Since each pixel has only one color, it appears in grayscale. We need to convert the Bayer pattern to RGB and then to YCrCb to produce a video with color output.

Color Conversion Software



Use this format as the output of your camera_loop() conversion pass

Describe the performance of your software-based color conversion (in terms of frames per second)



```
COM28 - PuTTY
Time: 1160012 us
Time: 1160035 us
Time: 1160024 us
Time: 1160030 us
Time: 1160032 us
Time: 1160017 us
Time: 1160031 us
Time: 1160027 us
Time: 1160024 us
Time: 1160012 us
Time: 1160028 us
Time: 1160022 us
Time: 1160017 us
Time: 1160025 us
Time: 1160030 us
Time: 1160024 us
Time: 1160026 us
Time: 1160013 us
Time: 1160013 us
Time: 1160016 us
Time: 1160017 us
Time: 1160024 us
Time: 1160019 us
```

FPS = Time per second / 1 sec reference = 1.16/1 = 1.16 frames per second

Describe your testing methodology

```
while(1) {
    Xil_DCacheInvalidateRange((UINTPTR)pS2MM_Mem, WIDTH * HEIGHT * sizeof(Xuint16));

    XTime_GetTime(&tStart);

    demosaic((Xuint16*)pS2MM_Mem, rgb_buffer);
    chromaConversion(rgb_buffer, (Xuint16*)pMM2S_Mem);

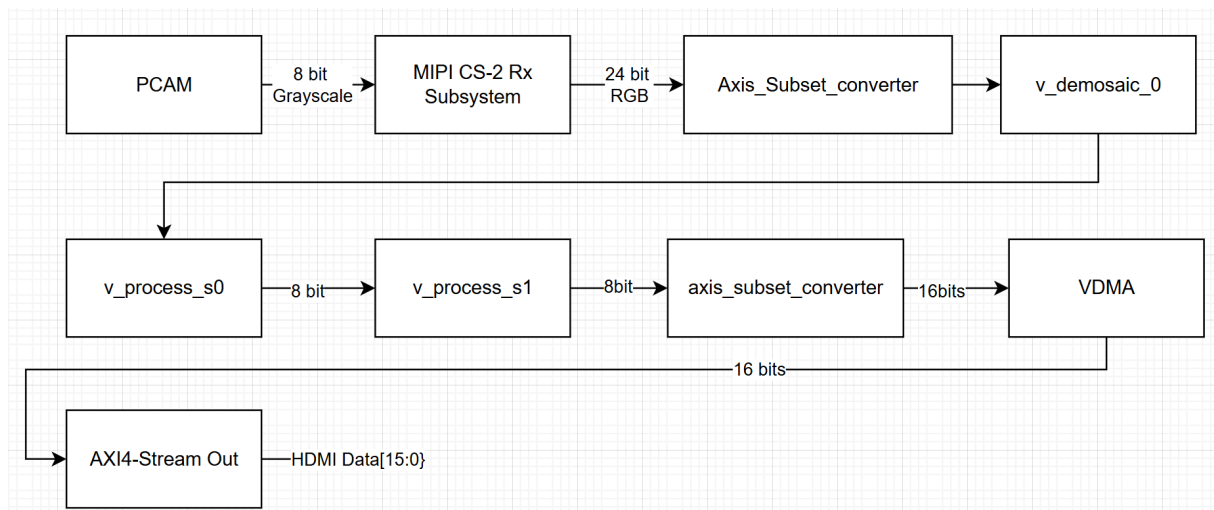
    XTime_GetTime(&tEnd);
    u64 elapsed_ticks = tEnd - tStart;
    duration_us = (uint32_t)((elapsed_ticks * 1000000) / COUNTS_PER_SECOND);

    xil_printf("Time: %u us", (unsigned int)duration_us);
    Xil_DCacheFlushRange((UINTPTR)pMM2S_Mem, WIDTH * HEIGHT * sizeof(Xuint16));
}
```

We created a timer in the code and measured how long it took to run the color algorithms.

Image Processing Pipeline

Include a diagram of this pipeline in your write-up, labeling the bit widths of the relevant signals.



Describe the performance of your image processing pipeline (in terms of frames per second), and how you measured it.

We were able to measure 30 frames per second on our camera system. We did this instead of using putty and a timer system like our software implementation, we instead used our phone to record over 1 second of our output and then counted the amount of frame changes in slow motion over a 1 second period and was able to roughly 30 frames per second.

Extra Credit

We were able to measure 30 frames per second on our camera system

- A video mode, which records and can replay up to 5 seconds of 1080p video. (5 bonus points).
- A digital zoom mode, which uses the up and down buttons to zoom in and out of the current scene. (5 bonus points).
- Various analog and digital adjustments for the gain, exposure, and other common user-configurable digital camera settings. (2 bonus points each: up to 6pts)